Aerodynamic Proof-of-Concept for a Dual-Regime Pressurized Morphing Wing Across Subsonic, Transonic, and Supersonic Regimes

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Abstract

This study presents a computational fluid dynamics (CFD) proof-of-concept evaluating a novel dual-regime pneumatic morphing wing operating from Mach 0.15 to Mach 2.5. The architecture features an internal rigid biconvex core surrounded by a flexible elastomeric skin that inflates into a high-lift supercritical profile for subsonic/transonic flight and evacuates flush against the core to form a razor-thin biconvex shape for supersonic flight. Static rigid-mesh simulations across 43 aerodynamic test cases demonstrate significant wave drag mitigation through the transonic dip zone (Mach 0.8–1.2) and an aerodynamic efficiency recovery to $(L/D)_{max} \approx 9.3$ at Mach 2.5. These static numerical results establish baseline feasibility prior to experimental laboratory wind-tunnel validation.

1. Morphing Architecture & CFD Methodology

1.1 Pneumatic Morphing Concept: The hybrid structural architecture combines an ultra-sharp, load-bearing rigid biconvex core with an outer flexible elastomeric composite skin. Pressurization of internal pneumatic chambers with Nitrogen expands the skin into a rounded, high-camber supercritical airfoil tailored for subsonic high-lift and transonic shock delay. Gas evacuation causes the skin to contract tightly against the rigid core, exposing a sharp biconvex profile to minimize supersonic wave drag.

1.2 Computational Setup: To establish aerodynamic proof-of-concept without the immediate overhead of dynamic mesh fluid-structure interaction (FSI), static rigid-mesh simulations were executed using OpenFOAM (*rhoSimpleFoam* and *rhoCentralFoam*). These simulations capture the discrete geometric boundary states (fully inflated supercritical vs. fully collapsed biconvex) across 43 flow conditions spanning Mach 0.15 to 2.5.

2. Results & Aerodynamic Performance

Lift curve slopes ($dC_L/d\alpha$) show linear subsonic behavior in the inflated supercritical state, achieving high peak efficiency ($(L/D)_{max} > 30$ at Mach 0.4). Transonic compressibility effects cause expected slope reductions between Mach 0.8 and 1.2.

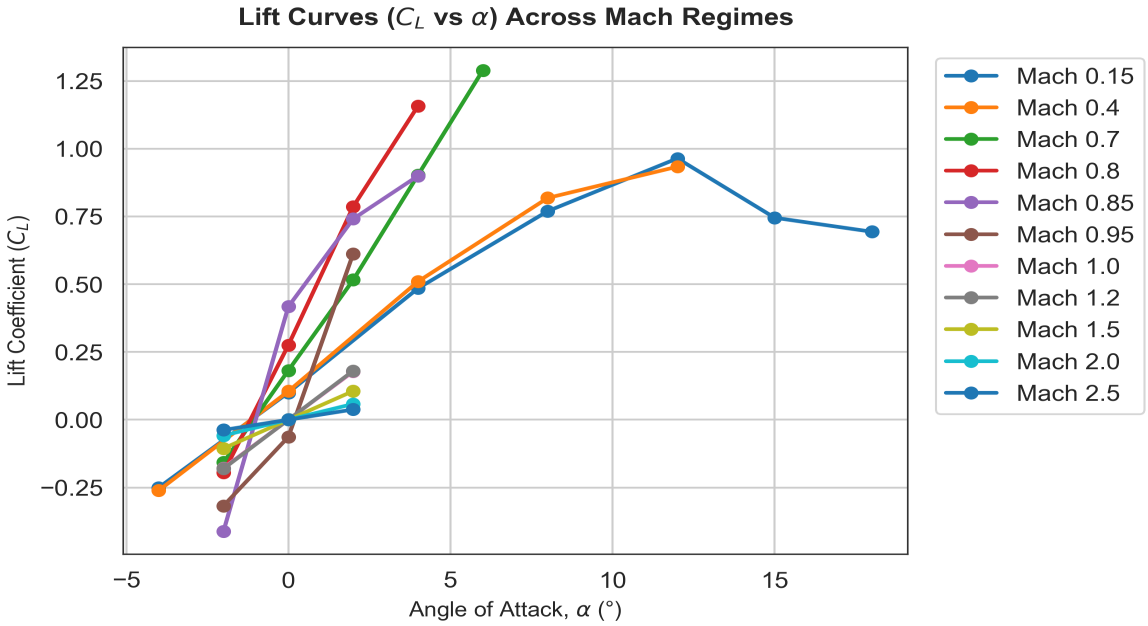


Figure 1: *Lift curves (C_L vs α) across subsonic, transonic, and supersonic speed regimes.*

Drag polars highlight zero-lift drag penalties (C_{D0}) during the transonic wave drag rise. Evacuating the skin into the thin biconvex profile for supersonic regimes suppresses frontal area and turns detached bow shocks into attached oblique shocks.

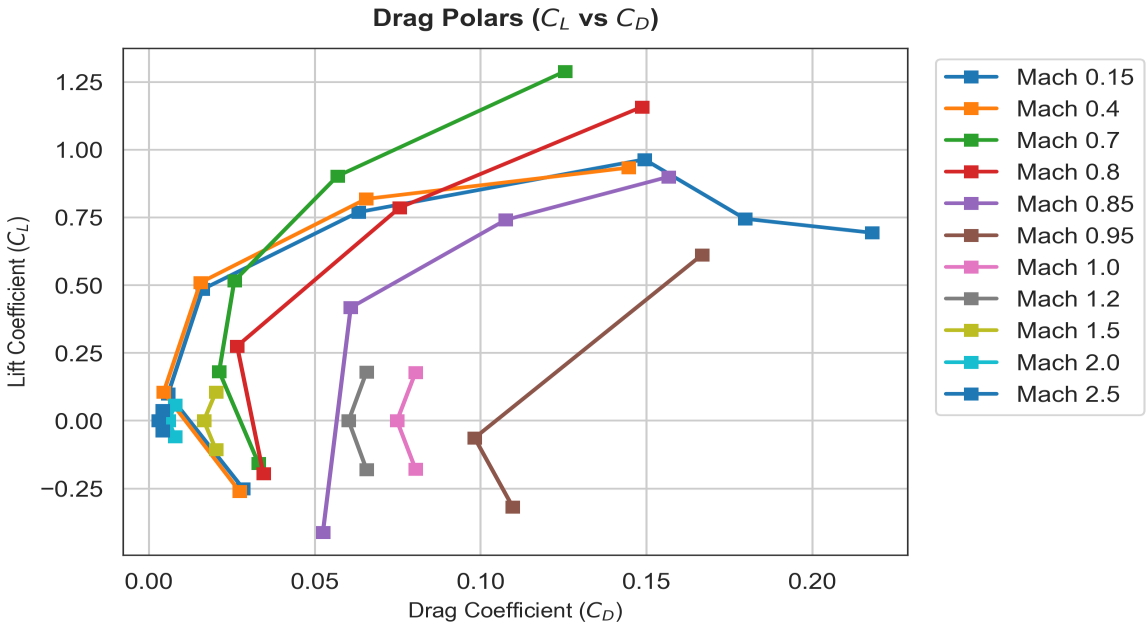


Figure 2: *Drag polars (C_L vs C_D) demonstrating wave drag transitions.*

The peak efficiency envelope highlights the transonic dip zone (Mach 0.8–1.2), where $(L/D)_{max}$ drops to localized minima before recovering to 9.3 at Mach 2.5 due to the sharp biconvex supersonic profile.

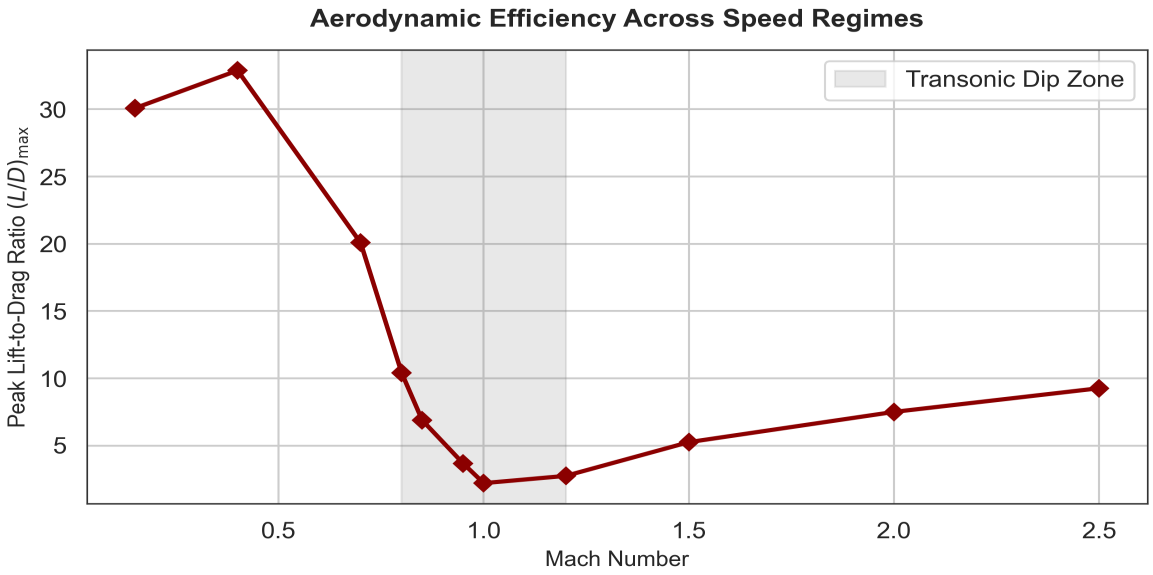


Figure 3: *Peak aerodynamic efficiency ($(L/D)_{max}$) across Mach numbers, highlighting the transonic dip zone.*

3. Conclusion & Experimental Roadmap

Static rigid-mesh CFD evaluations confirm the baseline aerodynamic proof-of-concept for the pressurized dual-regime morphing wing concept. The geometric transition effectively manages transonic wave drag and preserves supersonic efficiency. Following this publication, research will transition to experimental laboratory testing of a physical elastomeric prototype in a high-speed wind tunnel to validate dynamic pressure actuation, elastomeric skin deformation, and dynamic aeroelastic stability.